

## Qualification Pack



# Solar Manufacturing Junior Technician

QP Code: SGJ/Q4901

Version: 1.0

NSQF Level: 3

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri  
New Delhi - 110021 || email:prem@sscgj.in

## Qualification Pack

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## Qualification Pack

### SGJ/Q4901: Solar Manufacturing Junior Technician

#### Brief Job Description

Junior Technician Solar Manufacturing is responsible for basic preparation, handling and supply of materials for solar manufacturing. The job holder must adhere to all safety procedures and practice good housekeeping, safe operation of specific equipment, material or processes. This is an entry level position for solar manufacturing

#### Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it without any accidents so diligence and hardworking are desired attributes for individuals performing this role. He/She must also demonstrate strong work ethics, an ability to communicate courteously with co-workers and must be good with following instructions of the supervisor

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [SGJ/N4019: Explain and identify key components of Solar PV system](#)
2. [SGJ/N4020: Identify Different semi-conductor materials and types of solar cell](#)
3. [SGJ/N4021: Describe the Basics functions of Solar rooftop plant](#)
4. [SGJ/N4022: Maintain Personal Health & Safety for solar manufacturing](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                   |                  |
|-------------------|------------------|
| <b>Sector</b>     | Green Jobs       |
| <b>Sub-Sector</b> | Renewable Energy |
| <b>Occupation</b> | Assembler        |
| <b>Country</b>    | India            |
| <b>NSQF Level</b> | 3                |
| <b>Credits</b>    | 10               |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/8212.0400                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Minimum Educational Qualification &amp; Experience</b> | <p>9th grade pass (and pursuing continuous schooling in regular school.) with NA of experience<br/>OR<br/>8th grade pass (with two year of (NTC/ NAC) after 8th Grade Pass ) with 2 Years of experience<br/>OR<br/>Previous relevant Qualification of NSQF Level ( Level 2.5 ) with 1-2 Years of experience 1.5 years of relevant experience<br/>OR<br/>Previous relevant Qualification of NSQF Level (Level 2 ) with 3 Years of experience relevant experience</p> |
| <b>Minimum Level of Education for Training in School</b>  | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Minimum Job Entry Age</b>                              | 16 Years                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Next Review Date</b>                                   | 31/08/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>NSQC Approval Date</b>                                 | 31/08/2023                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Reference code on NQR</b>                              | QG-03-ES-00908-2023-V1-SCGJ                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>NQR Version</b>                                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Remarks:

Total 300 Hours duration including 140 hours theory, 100 hours practical and 30 hours of Employability Skills and 30 hours of OJT

## Qualification Pack

### SGJ/N4019: Explain and identify key components of Solar PV system

#### Description

This unit is about explain and identify different Renewable Energy Sources and its application

#### Scope

The scope covers the following :

- Identify key components of various renewable energy system
- Analyze important parameters of sunlight for Solar PV applications

#### Elements and Performance Criteria

##### *Identify key components of various renewable energy system*

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the Properties of renewable energy resources and its significance in power generation and illustrate the working of various renewable energy technology various charts and models
- PC2.** discuss the various Renewable energy system and its components and show the key components of the renewable energy system and outline their functions through plant schematic
- PC3.** discuss the selection of Renewable energy resources as per the suitable site locations
- PC4.** discuss the key components of wind turbine generator
- PC5.** discuss the key components of small hydro plant and bio gasifier
- PC6.** illustrate renewable energy technology through quiz and puzzles

##### *Analyze important parameters of sunlight for Solar PV applications.*

To be competent, the user/individual on the job must be able to:

- PC7.** discuss the properties of Sun light
- PC8.** discuss the Air Mass, Motion of the Sun
- PC9.** discuss the declination Angle, Elevation Angle Azimuth Angle and show the formation azimuth angle, declination angle
- PC10.** discuss about the sun's position
- PC11.** discuss about the Solar Radiation on a Tilted Surface
- PC12.** discuss the impact of temperature, GHI and DNI on power generation from solar PV and demonstrate the effect of temperature on Solar PV output

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process

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- KU3.** legal formalities applicable for the particular business
- KU4.** benefits to the company due to practice of these procedures
- KU5.** the importance of having an ethical and value based approach to governance
- KU6.** knowledge of processes involved in the business
- KU7.** country specific regulations for manufacturing and its importance
- KU8.** reporting procedure in case of deviations

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organisation's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Identify key components of various renewable energy system</i>                                                                                                                                     | <b>12</b>    | <b>13</b>       | -             | -          |
| <b>PC1.</b> discuss the Properties of renewable energy resources and its significance in power generation and illustrate the working of various renewable energy technology various charts and models | 3            | 5               | -             | -          |
| <b>PC2.</b> discuss the various Renewable energy system and its components and show the key components of the renewable energy system and outline their functions through plant schematic             | 3            | 3               | -             | -          |
| <b>PC3.</b> discuss the selection of Renewable energy resources as per the suitable site locations                                                                                                    | 2            | -               | -             | -          |
| <b>PC4.</b> discuss the key components of wind turbine generator                                                                                                                                      | 2            | -               | -             | -          |
| <b>PC5.</b> discuss the key components of small hydro plant and bio gasifier                                                                                                                          | 2            | -               | -             | -          |
| <b>PC6.</b> illustrate renewable energy technology through quiz and puzzles                                                                                                                           | -            | 5               | -             | -          |
| <i>Analyze important parameters of sunlight for Solar PV applications.</i>                                                                                                                            | <b>15</b>    | <b>10</b>       | -             | -          |
| <b>PC7.</b> discuss the properties of Sun light                                                                                                                                                       | 2            | -               | -             | -          |
| <b>PC8.</b> discuss the Air Mass, Motion of the Sun                                                                                                                                                   | 2            | -               | -             | -          |
| <b>PC9.</b> discuss the declination Angle, Elevation Angle Azimuth Angle and show the formation azimuth angle, declination angle                                                                      | 2            | 5               | -             | -          |
| <b>PC10.</b> discuss about the sun's position                                                                                                                                                         | 3            | -               | -             | -          |
| <b>PC11.</b> discuss about the Solar Radiation on a Tilted Surface                                                                                                                                    | 3            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> discuss the impact of temperature, GHI and DNI on power generation from solar PV and demonstrate the effect of temperature on Solar PV output | 3            | 5               | -             | -          |
| <b>NOS Total</b>                                                                                                                                           | <b>27</b>    | <b>23</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                        |
|----------------------------|--------------------------------------------------------|
| <b>NOS Code</b>            | SGJ/N4019                                              |
| <b>NOS Name</b>            | Explain and identify key components of Solar PV system |
| <b>Sector</b>              | Green Jobs                                             |
| <b>Sub-Sector</b>          | Renewable Energy                                       |
| <b>Occupation</b>          | Assembler                                              |
| <b>NSQF Level</b>          | 3                                                      |
| <b>Credits</b>             | 3                                                      |
| <b>Version</b>             | 1.0                                                    |
| <b>Last Reviewed Date</b>  | 31/08/2023                                             |
| <b>Next Review Date</b>    | 31/08/2026                                             |
| <b>NSQC Clearance Date</b> | 31/08/2023                                             |

## Qualification Pack

### SGJ/N4020: Identify Different semi-conductor materials and types of solar cell

#### Description

This unit is about to Identify different semi-conductor materials used in making solar cell.

#### Scope

The scope covers the following :

- Identify different semi-conductor materials used in making solar cells
- Identify different type of Silicon solar cells and their properties for solar manufacturing

#### Elements and Performance Criteria

##### *Identify different semi-conductor materials used in making solar cells*

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the brief introduction to semiconductor material and its properties
- PC2.** discuss the formation of p-n junction and demonstrate p-n junction formation by various means
- PC3.** discuss the various types of semiconductor material used in making solar cell
- PC4.** discuss the use of Gallium arsenide (GaAs), amorphous silicon and Cadmium Telluride (CdTe) Solar cell
- PC5.** discuss the properties of, GaAs, a-Si, CdTe, etc
- PC6.** show how to identify various elements used in solar cells from the periodic table
- PC7.** show the fabrication structure of solar cell and solar module
- PC8.** show generation of electricity from a solar cell by exposing it to light
- PC9.** show the effect of shadow on electricity generation from solar cell

##### *Identify different type of Silicon solar cells and their properties for solar manufacturing*

To be competent, the user/individual on the job must be able to:

- PC10.** discuss the various silicon based solar cell and show the working of solar cell through model and chart
- PC11.** discuss the type of silicon solar cell
- PC12.** discuss the multicrystalline and monocrystalline solar cell
- PC13.** discuss the properties of silicon
- PC14.** discuss in detail the working of silicon based solar module and show them silicon solar cell components and solar panel
- PC15.** discuss the advantage of silicon based solar cell
- PC16.** discuss the efficiency of silicon solar cell

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** the importance of having an ethical and value based approach to governance
- KU5.** knowledge of processes involved in the business
- KU6.** reporting procedure in case of deviations
- KU7.** Company's different department and concerned authority.
- KU8.** Company's Customer Support Policy

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organization's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Identify different semi-conductor materials used in making solar cells</i>                                        | <b>12</b>    | <b>16</b>       | -             | -          |
| <b>PC1.</b> discuss the brief introduction to semiconductor material and its properties                              | 2            | -               | -             | -          |
| <b>PC2.</b> discuss the formation of p-n junction and demonstrate p-n junction formation by various means            | 2            | 4               | -             | -          |
| <b>PC3.</b> discuss the various types of semiconductor material used in making solar cell                            | 2            | -               | -             | -          |
| <b>PC4.</b> discuss the use of Gallium arsenide (GaAs), amorphous silicon and Cadmium Telluride (CdTe) Solar cell    | 3            | -               | -             | -          |
| <b>PC5.</b> discuss the properties of, GaAs, a-Si, CdTe, etc                                                         | 3            | -               | -             | -          |
| <b>PC6.</b> show how to identify various elements used in solar cells from the periodic table                        | -            | 3               | -             | -          |
| <b>PC7.</b> show the fabrication structure of solar cell and solar module                                            | -            | 3               | -             | -          |
| <b>PC8.</b> show generation of electricity from a solar cell by exposing it to light                                 | -            | 3               | -             | -          |
| <b>PC9.</b> show the effect of shadow on electricity generation from solar cell                                      | -            | 3               | -             | -          |
| <i>Identify different type of Silicon solar cells and their properties for solar manufacturing</i>                   | <b>14</b>    | <b>8</b>        | -             | -          |
| <b>PC10.</b> discuss the various silicon based solar cell and show the working of solar cell through model and chart | 2            | 4               | -             | -          |
| <b>PC11.</b> discuss the type of silicon solar cell                                                                  | 2            | -               | -             | -          |
| <b>PC12.</b> discuss the multicrystalline and monocrystalline solar cell                                             | 2            | -               | -             | -          |
| <b>PC13.</b> discuss the properties of silicon                                                                       | 2            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC14.</b> discuss in detail the working of silicon based solar module and show them silicon solar cell components and solar panel | 2            | 4               | -             | -          |
| <b>PC15.</b> discuss the advantage of silicon based solar cell                                                                       | 2            | -               | -             | -          |
| <b>PC16.</b> discuss the efficiency of silicon solar cell                                                                            | 2            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                     | <b>26</b>    | <b>24</b>       | -             | -          |

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### National Occupational Standards (NOS) Parameters

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>NOS Code</b>            | SGJ/N4020                                                           |
| <b>NOS Name</b>            | Identify Different semi-conductor materials and types of solar cell |
| <b>Sector</b>              | Green Jobs                                                          |
| <b>Sub-Sector</b>          | Renewable Energy                                                    |
| <b>Occupation</b>          | Assembler                                                           |
| <b>NSQF Level</b>          | 3                                                                   |
| <b>Credits</b>             | 3                                                                   |
| <b>Version</b>             | 1.0                                                                 |
| <b>Last Reviewed Date</b>  | 31/08/2023                                                          |
| <b>Next Review Date</b>    | 31/08/2026                                                          |
| <b>NSQC Clearance Date</b> | 31/08/2023                                                          |

## Qualification Pack

### SGJ/N4021: Describe the Basics functions of Solar rooftop plant

#### Description

This unit is about basics functions of Solar rooftop plant

#### Scope

The scope covers the following :

- understand major components of solar rooftop plant
- tools and tackles required for installation of a small PV system

#### Elements and Performance Criteria

##### *Analyze the basic functions of Solar PV system*

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the solar small solar rooftop plant and demonstrate working model of small solar rooftop plants
- PC2.** discuss the major components of solar rooftop plant
- PC3.** discuss the electrical interconnection of small solar rooftop plant
- PC4.** discuss the function of a microgrid and demonstrate working model of micro grid
- PC5.** discuss the know-how of microgrid installation
- PC6.** discuss the advantage of using microgrid
- PC7.** discuss the tools and tackles required for installation of a small PV system
- PC8.** show connection of solar panel with charge controller and battery
- PC9.** perform measurement of Voltage and current of solar array and battery

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** the importance of having an ethical and value based approach to governance
- KU5.** knowledge of processes involved in the business
- KU6.** reporting procedure in case of deviations
- KU7.** Company's Customer Support Policy
- KU8.** Company's different department and concerned authority
- KU9.** Organization culture
- KU10.** Perform simple calculations to derive the power and energy received from solar radiation in a given area

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**KU11.** Mechanical and electrical features necessary for the long life of the PV Power Plant under a wide range of operating conditions.

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organization's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Analyze the basic functions of Solar PV system</i>                                                               | <b>30</b>    | <b>20</b>       | -             | -          |
| <b>PC1.</b> discuss the solar small solar rooftop plant and demonstrate working model of small solar rooftop plants | 4            | 5               | -             | -          |
| <b>PC2.</b> discuss the major components of solar rooftop plant                                                     | 5            | -               | -             | -          |
| <b>PC3.</b> discuss the electrical interconnection of small solar rooftop plant                                     | 5            | -               | -             | -          |
| <b>PC4.</b> discuss the function of a microgrid and demonstrate working model of micro grid                         | 4            | 5               | -             | -          |
| <b>PC5.</b> discuss the know-how of microgrid installation                                                          | 4            | -               | -             | -          |
| <b>PC6.</b> discuss the advantage of using microgrid                                                                | 4            | -               | -             | -          |
| <b>PC7.</b> discuss the tools and tackles required for installation of a small PV system                            | 4            | -               | -             | -          |
| <b>PC8.</b> show connection of solar panel with charge controller and battery                                       | -            | 5               | -             | -          |
| <b>PC9.</b> perform measurement of Voltage and current of solar array and battery                                   | -            | 5               | -             | -          |
| <b>NOS Total</b>                                                                                                    | <b>30</b>    | <b>20</b>       | -             | -          |

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### National Occupational Standards (NOS) Parameters

|                            |                                                      |
|----------------------------|------------------------------------------------------|
| <b>NOS Code</b>            | SGJ/N4021                                            |
| <b>NOS Name</b>            | Describe the Basics functions of Solar rooftop plant |
| <b>Sector</b>              | Green Jobs                                           |
| <b>Sub-Sector</b>          | Renewable Energy                                     |
| <b>Occupation</b>          | Assembler                                            |
| <b>NSQF Level</b>          | 3                                                    |
| <b>Credits</b>             | 1                                                    |
| <b>Version</b>             | 1.0                                                  |
| <b>Last Reviewed Date</b>  | 31/08/2023                                           |
| <b>Next Review Date</b>    | 31/08/2026                                           |
| <b>NSQC Clearance Date</b> | 31/08/2023                                           |

## Qualification Pack

### SGJ/N4022: Maintain Personal Health & Safety for solar manufacturing

#### Description

This unit is about maintain Personal Health & Safety for solar manufacturing

#### Scope

The scope covers the following :

- understand the requirements for safe work area
- Identify the hazards associated with solar manufacturing

#### Elements and Performance Criteria

##### *Maintain Personal Health & Safety for solar manufacturing*

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the requirements for safe work area
- PC2.** discuss the importance of administering first aid and demonstrate how to administer first aid
- PC3.** identify the personal protective equipment used for the specific purpose demonstrate the usage of personal protective equipment for ensuring safety during solar manufacturing.
- PC4.** discuss the hazards associated with solar manufacturing
- PC5.** discuss work safety procedures and instructions for solar manufacturing sites
- PC6.** discuss the importance of Occupational health & Safety standards and regulations for manufacturing Solar cells and modules
- PC7.** discuss good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** knowledge of processes involved in the business
- KU4.** the requirements for safe work area
- KU5.** the importance of administering first aid
- KU6.** the personal protective equipment used for the specific purpose
- KU7.** the hazards associated with solar manufacturing
- KU8.** work safety procedures and instructions for solar manufacturing sites
- KU9.** the importance of Occupational health & Safety standards and regulations for manufacturing Solar cells and modules
- KU10.** good housekeeping practices and infection control guidelines
- KU11.** how to administer first aid

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**KU12.** the usage of personal protective equipment for ensuring safety during solar manufacturing

**KU13.** good housekeeping and infection control & prevention practices

### Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** complete documentation work applicable to the role

**GS2.** read and write in vernacular/English language

**GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature

**GS4.** express statements with required clarity

**GS5.** participate in and understand the main points of simple discussions

**GS6.** follow organization's rule-based decision making process

**GS7.** work constructively and collaboratively with others

**GS8.** follow organization's code of conduct

**GS9.** approach relevant authority when required

**GS10.** ask questions for better understanding

**GS11.** communicate with team members and colleagues on the significance of greening of jobs

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Maintain Personal Health &amp; Safety for solar manufacturing</i>                                                                                                                        | <b>35</b>    | <b>15</b>       | -             | -          |
| <b>PC1.</b> discuss the requirements for safe work area                                                                                                                                     | 5            | -               | -             | -          |
| <b>PC2.</b> discuss the importance of administering first aid and demonstrate how to administer first aid                                                                                   | 5            | -               | -             | -          |
| <b>PC3.</b> identify the personal protective equipment used for the specific purpose demonstrate the usage of personal protective equipment for ensuring safety during solar manufacturing. | 5            | -               | -             | -          |
| <b>PC4.</b> discuss the hazards associated with solar manufacturing                                                                                                                         | 5            | -               | -             | -          |
| <b>PC5.</b> discuss work safety procedures and instructions for solar manufacturing sites                                                                                                   | 5            | -               | -             | -          |
| <b>PC6.</b> discuss the importance of Occupational health & Safety standards and regulations for manufacturing Solar cells and modules                                                      | 5            | -               | -             | -          |
| <b>PC7.</b> discuss good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices                             | 5            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                            | <b>35</b>    | <b>15</b>       | -             | -          |

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### National Occupational Standards (NOS) Parameters

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| <b>NOS Code</b>            | SGJ/N4022                                                 |
| <b>NOS Name</b>            | Maintain Personal Health & Safety for solar manufacturing |
| <b>Sector</b>              | Green Jobs                                                |
| <b>Sub-Sector</b>          | Renewable Energy                                          |
| <b>Occupation</b>          | Assembler                                                 |
| <b>NSQF Level</b>          | 4                                                         |
| <b>Credits</b>             | 1                                                         |
| <b>Version</b>             | 1.0                                                       |
| <b>Last Reviewed Date</b>  | 31/08/2023                                                |
| <b>Next Review Date</b>    | 31/08/2026                                                |
| <b>NSQC Clearance Date</b> | 31/08/2023                                                |

## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team

## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services



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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

## Qualification Pack

### Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|-------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| SGJ/N4019.Explain and identify key components of Solar PV system              | 27           | 23              | 0             | 0          | 50          | 20         |
| SGJ/N4020.Identify Different semi-conductor materials and types of solar cell | 26           | 24              | 0             | 0          | 50          | 20         |
| SGJ/N4021.Describe the Basics functions of Solar rooftop plant                | 30           | 20              | 0             | 0          | 50          | 20         |
| SGJ/N4022.Maintain Personal Health & Safety for solar manufacturing           | 35           | 15              | 0             | 0          | 50          | 20         |
| DGT/VSQ/N0101.Employability Skills (30 Hours)                                 | 20           | 30              | -             | -          | 50          | 20         |
| <b>Total</b>                                                                  | <b>138</b>   | <b>112</b>      | <b>-</b>      | <b>-</b>   | <b>250</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |